

CLINICAL DATA MANAGEMENT PROJECT

CLINICAL TRIAL DATA REVIEW & QUERY MANAGEMENT

(Self-Directed Clinical Data Management Practice Project)

Study ID	CDM-MIG-001
Study Phase	Phase II
Study Design	Randomized, Double-Blind, Placebo-Controlled
Indication	Migraine
Data Management Tool	Microsoft Excel

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1.PROJECT OVERVIEW

This self-directed project was designed to simulate key Clinical Data Management (CDM) activities involved in the review and cleaning of clinical trial data.

A fictional Phase II randomized, double-blind, placebo-controlled clinical trial in patients with migraine was used as the basis for the exercise. Simulated clinical trial data were created across multiple data domains, followed by predefined validation checks, discrepancy identification, query management, data correction, cleaning documentation, and final quality control.

The project was performed using Microsoft Excel to demonstrate a structured clinical data review workflow in the absence of access to a commercial Electronic Data Capture (EDC) system.

Project Objectives

- Review simulated clinical trial data for completeness, consistency, and validity.
- Apply predefined data validation rules across multiple clinical data domains.
- Identify missing, inconsistent, or potentially erroneous data.
- Create and manage data queries based on identified discrepancies.
- Document simulated site responses and track query resolution.
- Perform data corrections following documented query resolution.
- Maintain a data cleaning log and conduct final quality control.
- Demonstrate understanding of the clinical data management workflow from data review through QC.

Project Scope

The project covered the following clinical data domains:

- Demographics
- Medical History
- Efficacy
- Adverse Events
- Vital Signs
- Laboratory Data
- Treatment Data

2.STUDY DESIGN AND PROTOCOL

2.1 Study Overview

Parameter	Description
Study ID	CDM-MIG-001
Study Title	A Phase II Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Drug X in Patients With Migraine
Study Phase	Phase II
Study Design	Randomized, double-blind, placebo-controlled, parallel-group
Study Population	Adults aged 18–65 years diagnosed with migraine
Planned Subjects	20
Treatment Groups	Drug X / Placebo
Study Duration	12 weeks
Data Management Tool	Microsoft Excel

2.2 Study Visits

Visit	Study Time Point	Key Data Collected
Visit 1	Screening	Demographics, medical history, laboratory data
Visit 2	Baseline / Day 1	Efficacy, vital signs, laboratory data, treatment
Visit 3	Week 4	Efficacy, adverse events, vital signs, treatment
Visit 4	Week 8	Efficacy, adverse events, vital signs, treatment
Visit 5	Week 12 / End of Treatment	Efficacy, adverse events, vital signs, laboratory data, treatment

2.3 Data Domains

The simulated dataset consisted of seven primary data domains:

1. Demographics
2. Medical History
3. Efficacy
4. Adverse Events
5. Vital Signs
6. Laboratory Data
7. Treatment

2.4 Data Management Objectives

The project focused on assessing clinical trial data for:

- Completeness of required data fields
- Validity of values against predefined ranges
- Consistency within and across data domains
- Identification and documentation of discrepancies
- Query generation and resolution tracking
- Data cleaning and final quality control

3. CRF DESIGN AND DATA STRUCTURE

3.1 CRF Design Approach

A simulated Case Report Form (CRF) structure was developed before generating the clinical trial dataset. The CRF design defined the variables to be collected, data types, required fields, permitted values or ranges, study visits, and descriptions for each variable.

The CRF structure was designed to support subsequent data validation and discrepancy identification.

3.2 CRF Domains

The CRF consisted of seven data domains:

Domain	Examples of Variables
Demographics	Subject ID, age, sex, height, weight, screening date
Medical History	Migraine diagnosis, migraine duration, medical conditions, previous treatment
Efficacy	Monthly migraine days, headache severity, rescue medication
Adverse Events	AE term, onset date, resolution date, severity, seriousness
Vital Signs	Blood pressure, heart rate, temperature, respiratory rate
Laboratory Data	Hemoglobin, WBC, platelets, ALT
Treatment	Treatment assignment, dose, treatment dates, compliance

3.3 Data Definition Components

Each CRF variable was defined using the following components:

- **Field Name** – clinical variable represented in the CRF.
- **Variable Name** – standardized variable identifier.
- **Data Type** – text, numeric, categorical, or date.
- **Required Status** – identifies mandatory and conditional fields.
- **Allowed Values / Range** – predefined acceptable values or numerical limits.
- **Visit** – study time point at which the variable is collected.
- **Description** – purpose and meaning of the variable.

3.4 Example CRF Variables

Field	Variable	Type	Required	Validation Rule
Age	AGE	Numeric	Yes	18–65 years
Sex	SEX	Categorical	Yes	Male / Female
Weight	WEIGHT	Numeric	Yes	30–200 kg
Migraine Days	MIGRAINE_DAYS	Numeric	Yes	0–31
Headache Severity	HEADACHE_SEVERITY	Numeric	Yes	0–10
Treatment Assignment	TREATMENT	Categorical	Yes	Drug X / Placebo
Compliance	COMPLIANCE	Numeric	Yes	0–100%

3.5 Purpose of the CRF Design

The CRF structure served as the reference for generating the simulated clinical trial dataset and establishing validation rules. Defining expected data characteristics before data review allowed discrepancies to be assessed against predefined requirements rather than subjective judgment.

4. RAW DATA AND DATA PREPARATION

4.1 Simulated Clinical Trial Dataset

A simulated clinical trial dataset was created based on the predefined CRF structure and study protocol. The dataset represented 20 subjects enrolled in the fictional Phase II migraine study and included multiple visits and clinical data domains.

The purpose of the dataset was to provide realistic practice data for clinical data review, validation, discrepancy identification, query management, and data cleaning.

4.2 Dataset Structure

Data Domain	Records	Key Data Elements
Demographics	20	Age, sex, height, weight, screening date
Medical History	20	Diagnosis, migraine duration, medical conditions, previous treatment
Efficacy	80	Migraine days, headache severity, rescue medication
Adverse Events	20	AE occurrence, term, dates, severity, seriousness
Vital Signs	80	Blood pressure, heart rate, temperature, respiratory rate
Laboratory Data	60	Hemoglobin, WBC, platelets, ALT
Treatment	80	Treatment assignment, dose, dates, compliance

4.3 Data Preparation

The simulated dataset was structured in separate Excel worksheets according to the defined CRF domains. Each record was associated with a unique subject identifier to support subject-level tracking and cross-domain review.

The data were intentionally populated with a small number of simulated discrepancies to provide a controlled environment for practicing clinical data validation and query management. The discrepancies were not corrected during initial data entry and were identified through predefined validation checks.

4.4 Data Review Approach

The raw dataset was reviewed against the predefined CRF requirements and validation rules. The review considered:

- Required field completeness
- Permitted categorical values
- Numerical range checks
- Conditional field consistency
- Cross-visit consistency
- Cross-domain consistency
- Date and event consistency

The original raw data were retained during the initial review. Corrections were performed only after the corresponding simulated queries had been resolved and documented.

5. DATA VALIDATION AND DISCREPANCY IDENTIFICATION

5.1 Validation Approach

The simulated clinical trial data were reviewed using predefined validation rules derived from the study protocol and CRF design.

Validation checks were performed across demographic, medical history, efficacy, safety, laboratory, vital signs, and treatment data. The objective was to identify missing, invalid, inconsistent, or potentially erroneous values requiring further clarification.

5.2 Validation Checks Performed

The following types of checks were performed:

- **Range checks** – numerical values were compared with predefined acceptable ranges.
- **Completeness checks** – required fields were reviewed for missing values.
- **Categorical checks** – values were reviewed against predefined allowed values.
- **Conditional checks** – related fields were assessed for logical consistency.
- **Cross-visit checks** – subject data were compared across study visits.
- **Cross-domain checks** – related information was compared between different data domains.
- **Date consistency checks** – applicable dates were reviewed for logical sequence.

5.3 Identified Discrepancies

Four discrepancies requiring clarification were identified during the validation process.

Check ID	Subject	Domain	Variable	Finding
VAL001	SUBJ006	Demographics	Age	Age recorded as 17 years, below the protocol range of 18–65 years
VAL002	SUBJ007	Demographics	Sex	Required sex field was missing
VAL003	SUBJ008	Demographics	Weight	Weight recorded as 350 kg, outside the predefined range of 30–200 kg
VAL004	SUBJ010	Treatment	Treatment Assignment	Treatment assignment changed from Placebo to Drug X and then returned to Placebo

5.4 Validation Outcome

The identified discrepancies were documented in the Data Validation Log and were not directly corrected during the initial review. Each finding was converted into a corresponding data query for clarification and resolution.

Other reviewed domains and validation checks did not identify additional discrepancies requiring queries.

6. QUERY MANAGEMENT AND RESOLUTION

6.1 Query Management Approach

Following discrepancy identification, each finding was documented as a data query. Queries were written to clearly describe the identified issue and request confirmation or clarification of the clinical data.

The query process followed the sequence:

Discrepancy Identified → Query Raised → Site Response → Query Resolved → Data Corrected

6.2 Query Summary

Query ID	Subject	Variable	Query Type	Status
QRY001	SUBJ006	Age	Out-of-range value	Resolved
QRY002	SUBJ007	Sex	Missing required value	Resolved
QRY003	SUBJ008	Weight	Out-of-range value	Resolved
QRY004	SUBJ010	Treatment Assignment	Inconsistent treatment assignment	Resolved

6.3 Query Resolution

All four queries received simulated site responses that provided clarification for the identified discrepancies.

Query ID	Original Finding	Site Response / Clarification	Resolution
QRY001	Age recorded as 17 years	Site confirmed the subject's correct age was 27 years and that the original value was entered incorrectly.	Age corrected to 27 years
QRY002	Sex field was missing	Site confirmed the subject's sex as Female.	Sex updated to Female
QRY003	Weight recorded as 350 kg	Site confirmed the correct weight was 75 kg and that the original value was a data-entry error.	Weight corrected to 75 kg
QRY004	Treatment assignment was inconsistent	Site confirmed the Week 4 Drug X entry was a data-entry error and the correct assignment was Placebo.	Treatment assignment corrected to Placebo

6.4 Query Closure

After receiving and documenting the simulated site responses, each query was marked as resolved. The corresponding corrections were then documented in the Data Cleaning Log.

No query was closed without a documented resolution.

7.DATA CLEANING AND QUALITY CONTROL

7.1 Data Cleaning Approach

After query resolution, the confirmed corrections were applied to the relevant clinical trial data. Each correction was documented in the Data Cleaning Log with the original value, corrected value, and reason for correction.

The cleaning process maintained traceability between the original discrepancy, query resolution, and final corrected value.

7.2 Data Corrections

Cleaning ID	Subject	Variable	Original Value	Corrected Value	Reason
CLN001	SUBJ006	Age	17	27	Site confirmed the correct age and the original value was a data-entry error.
CLN002	SUBJ007	Sex	Missing	Female	Site confirmed the correct sex and the missing value was updated.
CLN003	SUBJ008	Weight	350 kg	75 kg	Site confirmed a data-entry error and provided the correct weight.
CLN004	SUBJ010	Treatment Assignment	Drug X	Placebo	Site confirmed the Week 4 Drug X entry was a data-entry error.

7.3 Quality Control Review

A final QC review was performed after the corrections were applied. Each corrected record was checked against the corresponding site response and resolved query.

QC ID	Subject	Variable	QC Check	Result
QC001	SUBJ006	Age	Corrected age matched the site response and resolved query	Pass
QC002	SUBJ007	Sex	Corrected sex matched the site response and resolved query	Pass
QC003	SUBJ008	Weight	Corrected weight matched the site response and resolved query	Pass
QC004	SUBJ010	Treatment Assignment	Corrected treatment assignment matched the site response and resolved query	Pass

7.4 Final QC Outcome

All four corrected records passed the final QC review. The review confirmed that the documented corrections were consistent with the corresponding query resolutions.

The completed workflow demonstrated traceability from discrepancy identification through query management, data correction, and final quality control.

8.FINAL DATA REVIEW AND OUTCOME

8.1 Final Data Review

Following completion of query resolution, data correction, and QC, a final review of the simulated clinical trial dataset was performed.

The review covered all seven clinical data domains and focused on completeness, validity, consistency, and the successful application of documented corrections.

Data Domain	Final Review Status
Demographics	Pass
Medical History	Pass
Efficacy	Pass
Adverse Events	Pass
Vital Signs	Pass
Laboratory Data	Pass
Treatment	Pass

8.2 Query and QC Outcome

A total of **4 discrepancies** were identified during the data validation process.

- **4 queries raised**
- **4 queries resolved**
- **4 data corrections documented**
- **4 QC checks performed**
- **4 QC checks passed**

This resulted in a completed discrepancy-to-resolution workflow with documented traceability.

8.3 Project Outcome

The project successfully simulated a structured clinical data management workflow using a controlled clinical trial dataset.

The completed workflow covered:

CRF Design → Data Generation → Data Validation → Discrepancy Identification → Query Management → Site Response → Data Correction → QC → Final Data Review

The exercise provided practical exposure to the documentation and review activities involved in clinical data management while maintaining a clear distinction between simulated practice and professional clinical trial experience.

8.4 Key Learning Outcomes

Through this project, the following areas were practiced:

- Clinical data review
- CRF-based data structure understanding
- Data validation
- Range and consistency checks
- Query identification and documentation
- Query resolution tracking
- Data cleaning
- Correction traceability
- Quality control review
- Clinical trial data documentation

9.SKILLS DEMONSTRATED AND TOOLS USED

9.1 Clinical Data Management Skills

The project provided practice in the following Clinical Data Management activities:

- Understanding of CRF structure and clinical data fields
- Clinical data review and validation
- Identification of missing and inconsistent data
- Range and edit-check based review
- Query generation and documentation
- Query status and resolution tracking
- Data cleaning and correction documentation
- Traceability between queries and corrections
- Final QC review
- Clinical trial data documentation

9.2 Data Review Concepts Practiced

The project included practical application of the following concepts:

Area	Application in Project
CRF Review	Reviewed variables, data types, required fields, and validation requirements
Data Validation	Applied predefined range, completeness, categorical, and consistency checks
Query Management	Documented discrepancies and tracked simulated site responses
Data Cleaning	Applied confirmed corrections and maintained correction history
QC	Verified corrected values against documented query resolutions
Traceability	Linked validation findings, queries, corrections, and QC results

9.3 Tools Used

Microsoft Excel

Excel was used to create and manage the simulated clinical trial dataset and supporting CDM documentation, including:

- CRF design
- Raw clinical data
- Data validation log
- Query log
- Data cleaning log
- QC log

Excel functions and features were used for structured data entry, filtering, sorting, review, and organization of clinical data.

Microsoft Word

Word was used to prepare the final project report and document the overall CDM workflow, findings, corrections, QC results, and project outcomes.

9.4 Professional Skills Practiced

The project also strengthened:

- Attention to detail
- Structured documentation
- Data accuracy and consistency review
- Analytical thinking
- Problem identification
- Documentation traceability
- Systematic quality checking

10.PROJECT LIMITATIONS

This project was designed as a self-directed clinical data management practice exercise using simulated clinical trial data. The following limitations should be considered:

- The dataset was simulated and was not obtained from an actual clinical trial.
- No real patient information or protected health information was used.
- Microsoft Excel was used instead of a commercial EDC or clinical data management system.
- Site responses and query resolutions were simulated for practice purposes.
- The project did not involve actual study-site communication or real clinical trial monitoring.
- The validation rules were developed specifically for the simulated study and do not represent a complete production clinical database validation specification.

Despite these limitations, the project provided a controlled environment for practicing the sequence of clinical data review, discrepancy identification, query management, data cleaning, and quality control.

11.CONCLUSION

This self-directed project demonstrated a structured approach to clinical data management using a simulated Phase II migraine clinical trial dataset.

The project covered the complete data review workflow from CRF design and data preparation through validation, query management, data correction, quality control, and final review. Four simulated discrepancies were identified, queried, resolved, corrected, and successfully verified through QC.

The exercise strengthened practical understanding of clinical data review, query management, data cleaning, documentation, and quality control while maintaining a clear distinction between simulated practice and professional clinical trial experience.

The completed project and supporting Excel files provide documented evidence of the CDM concepts and activities practiced during the exercise.

12.PROJECT DELIVERABLES

The following files were created as part of the self-directed Clinical Data Management practice project:

No.	Deliverable	Purpose
01	Study Protocol	Defines the simulated study design, population, visits, and data requirements
02	CRF Design	Defines the clinical data fields, data types, and validation requirements
03	Clinical Trial Raw Data	Contains the simulated subject-level clinical trial data
04	Data Validation Log	Documents validation checks and identified discrepancies
05	Query Log	Documents queries, site responses, and resolution status
06	Data Cleaning Log	Records confirmed corrections and reasons for correction
07	QC Log	Documents final quality control checks of corrected data
08	Final Project Report	Summarizes the complete CDM workflow, findings, and outcomes

Project Workflow

The deliverables were developed as a connected workflow rather than as independent files:

Protocol → CRF → Raw Data → Validation Log → Query Log → Cleaning Log → QC Log → Final Report

Each stage provided input or documentation for the subsequent stage, supporting traceability throughout the simulated data management process.

Project Classification

Project Type: Self-Directed Practice Project

Data Type: Simulated Clinical Trial Data

System Used: Microsoft Excel

Reporting Tool: Microsoft Word

This project should be presented as a **practice project** and not as professional clinical trial work experience.